

THE MACROECONOMIC EFFECTS OF LIFTING BARRIERS TO MIGRATION

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STARK URBAN-RURAL GAP IN LOW-INCOME COUNTRIES



Pictures show a rural Kenyan home and Nairobi, the capital city of Kenya. Source: Jocelyn Diaz (left) and Catalin Marin (right).

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- urbanization remains limited
- growing randomized controlled trial evidence on why!

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which interventions are most effective?

- unclear because experiments have
 - different environment and sampling
 - different intervention design and measurement
 - models tailored to specific experiment

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- estimate to match moments from Kenyan data
- reproduce **several prominent experiments** in Kenya within a single environment
 - approximate sampling, intervention, measurement of each
 - unconditional cash transfers (*Miner, 2025; Egger et al., 2022*)
 - unemployment insurance (*Miner, 2025*)
 - information treatments (*Barnett-Howell et al., 2025*)

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- use the estimated model to
 1. assess **long-run effects** of each intervention beyond the experimental horizon
 2. evaluate **macro effects** of scaling up each intervention nationally
 3. **decompose** the urban-rural income gap into contributions from different frictions

PREVIEW: MODEL

heterogeneous agent general equilibrium Aiyagari model

- **households** choose consumption, savings, and location each period

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 - **borrowing constraint:** limits ability to finance the move
 - **urban unemployment risk:** leaves room for UI
 - **information friction:** villagers underestimate urban earnings
 - **migration disutility:** inexperienced migrants discount urban consumption
 - **migration cost:** cost of moving, picks up all other barriers to migration

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- solve for **stationary general equilibrium**

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- model reproduces **treatment effects** fairly well
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 - information interventions move urbanization modestly
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- **national rollout** of unemployment insurance is most promising
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 - unemployment insurance highly effective at increasing urbanization
- decomposing the **urban-rural income gap**:
 - information friction least important
 - financial friction explains much of the current migration, small part of the gap
 - residual migration costs explain most of the gap

EXPERIMENTS BACKGROUND

OVERVIEW OF EXPERIMENTS

- several experiments conducted in Kenya, targeting different barriers to migration
- **cash transfer:**
 - *Egger et al. (2022)*
 - *Miner (2025)*
- **unemployment insurance:** *Miner (2025)*
- **information provision:** *Barnett-Howell et al. (2025)*

CASH TRANSFER: EGGER ET AL. (2022)

Egger, Haushofer, Miguel, Niehaus, Walker (2022)

- **sample:** 10,500 poor households across 653 rural villages
 - randomly sampled from 653 villages within 1 county
- **intervention:** one-time huge unconditional cash transfer
 - 75% of median annual household income
 - 16% of annual village GDP
- **findings:**
 - expenditure $\uparrow$ 12%
 - no migration effect
 - large local spillovers: consumption $\uparrow$ similarly among treated and untreated
 - no change in local labor supply
 - local firms: sales $\uparrow$, wages $\uparrow$
- **requirements for model:**
 - financial frictions
 - mechanism for large local spillovers through demand channel

CASH TRANSFER: MINER (2025)

Miner (2025)

- **sample:** 700 young male workers
 - randomly sampled from villages within 1 county
- **intervention:** one-time unconditional cash transfer similar to maximum UI benefits in UI arm
- **findings:**
 - migration rate $\uparrow$ 5 p.p.
 - no significant income change

UNEMPLOYMENT INSURANCE: MINER (2025)

Miner (2025)

- **sample:** 700 young male workers
 - randomly sampled from villages within 1 county
- **intervention:** offer daily UI in Nairobi
 - ~ daily urban wage
 - up to 14 days
- **findings:**
 - migration rate $\uparrow$ 11 p.p.
 - no significant income change
- **requirements for model:**
 - unemployment risk
 - potential unemployment benefit

INFORMATION PROVISION: BARNETT-HOWELL ET AL. (2025)

Barnett-Howell, Baseler, Ginn, Gordeev (2025)

- **sample:** 17,000 rural households
 - representatively sampled from 5 rural counties
- **intervention:** information about Nairobi income, labor market
- treatment arms:
 - **information:** individual delivery
 - **mentor:** individual + mentor pairing
 - **group:** delivery in village-wide meeting with group discussion

INFORMATION PROVISION: BARNETT-HOWELL ET AL. (2025)

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- **findings:**
 - villagers underestimate conditional urban income by 44%
 - treatment → migration rate ↑ 2 p.p.
 - individual/mentor treatment → income ↑ 9%
 - group treatment → no income change
 - group treatment primarily engages experienced migrants, who have low marginal returns (because of low migration cost)
 - large economic spillovers within village
- **requirements for model:**
 - information friction
 - heterogeneous migration experience among villagers affecting migration cost

MODEL

MODEL OVERVIEW

Aiyagari-style dynamic heterogeneous agent model with migration

REGIONS

- Rural (R) and Urban (U)

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- location l
- urban experience x
- employment status e
- assets a

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FRICTIONS

- borrowing constraint: $a \geq \underline{a}$
- information friction:
 - rurals perceive $\tilde{w}_U = \gamma w_U$
- urban unemployment risk:
 - job separation λ_s
 - job finding λ_f
- migration costs: pay m_c when moving
- migration disutility:
 - the inexperienced discount urban consumption by m_u

PREFERENCES AND EXPERIENCE

- CRRA utility over effective consumption:

$$u(c, l, x) = \frac{\tilde{c}^{1-\sigma} - 1}{1-\sigma}, \quad \tilde{c} = \frac{c}{1 + m_u \cdot \mathbf{1}(l = U, x = 0)}$$

- m_u : consumption-equivalent disutility for inexperienced urban migrants
 - inexperienced migrants in urban region discount consumption

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 - inexperienced migrants in urban region discount consumption
- urban experience $x \in \{0, 1\}$:
 - gain experience ($x' = 1$) with prob. π_x per period in urban, stay experienced
 - lose experience ($x' = 0$) with prob. π_{-x} per period in rural, stay inexperienced
 - as in *Lagakos, Mobarak, Waugh (2023)*

INCOME, PRODUCTIVITY, AND EMPLOYMENT

budget constraint: $c + a' + \mathbf{1}(l = R, l' = U) \cdot m_c = (1 + r)a + y(e, l, z_R, z_U)$

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EMPLOYMENT

- urban: stochastic employment e
 - $e = 1 \rightarrow 0$: job separation rate λ_s
 - $e = 0 \rightarrow 1$: job finding rate λ_f
- rural: always employed ($e = 1$)

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- pay m_c when moving $R \rightarrow U$

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BIVARIATE AR(1) PRODUCTIVITY PROCESS

$$\log z'_R = \rho_R \log z_R + \epsilon'_R$$

$$\log z'_U = \rho_U \log z_U + \epsilon'_U$$

$$(\epsilon'_R, \epsilon'_U) \sim N(\mathbf{0}, \Sigma)$$

- ρ_ϵ : cross-correlation of shocks

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- ρ_ϵ : cross-correlation of shocks

ASSETS

- save in one-period risk-free bond a'
 - return r
- borrowing constraint: $a' \geq \underline{a}$

INFORMATION FRICTION

- rural households perceive urban wage as $\tilde{w}_U = \gamma w_U$, where $\gamma \in (0, 1]$
 - $\gamma < 1$: rural households underestimate potential urban income
 - perceived income:

$$\tilde{y}(e, l, z_R, z_U, \gamma) = \begin{cases} w_R z_R & \text{if } l = R \\ \gamma w_U z_U & \text{if } l = U \text{ and } e = 1 \\ b & \text{if } l = U \text{ and } e = 0 \end{cases}$$

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- as in *Barnett-Howell et al. (2025)*
- rural→urban migrants are surprised by true urban wage
- urban households don't anticipate forgetting the true urban wage
 - but revert to misperception if they return to rural

RECURSIVE FORMULATION: RURAL HOUSEHOLD

rural household stays or migrates to urban:

$$V_R(a, z_R, z_U, x, \gamma) = \max \{ V_{R \rightarrow R}, V_{R \rightarrow U} \}$$

STAY IN RURAL

$$V_{R \rightarrow R} = \max_{a' \geq \underline{a}} u(c, R, x) + \beta \mathbb{E}[V_R(a', z'_R, z'_U, x', \gamma)]$$

$$c + a' = (1 + r)a + \tilde{y}(1, R, z_R, z_U, \gamma)$$

MIGRATE TO URBAN

$$V_{R \rightarrow U} = \max_{a' \geq \underline{a}} u(c, R, x) + \beta \mathbb{E}[V_U(a', z'_R, z'_U, x', e', \gamma)]$$

$$c + a' + m_c = (1 + r)a + \tilde{y}(1, R, z_R, z_U, \gamma)$$

RECURSIVE FORMULATION: URBAN HOUSEHOLD

urban household stays or migrates to rural:

$$V_U(a, z_R, z_U, x, e, \gamma) = \max \{V_{U \rightarrow U}, V_{U \rightarrow R}\}$$

STAY IN URBAN

$$V_{U \rightarrow U} = \max_{a' \geq \underline{a}} u(c, U, x) + \beta \mathbb{E}[V_U(a', z'_R, z'_U, x', e', \gamma)]$$

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MIGRATE TO RURAL

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URBAN HOUSEHOLDS ARE MYOPIC IN TWO WAYS

1. **Income beliefs:** while in U , households use V with $\gamma = 1$
 - do not anticipate forgetting the true urban wage after returning to R
 - otherwise everyone becomes well-informed in stationary equilibrium

URBAN HOUSEHOLDS ARE MYOPIC IN TWO WAYS

1. **Income beliefs:** while in U , households use V with $\gamma = 1$
 - do not anticipate forgetting the true urban wage after returning to R
 - otherwise everyone becomes well-informed in stationary equilibrium
2. **Migration cost beliefs:** while in U , households use V with $m_c = 0$
 - do not anticipate facing migration cost after returning to R for repeat $R \rightarrow U$ moves
 - allows the stationary equilibrium to hit URPG, urban share moments

PRODUCTION AND GENERAL EQUILIBRIUM

Production

- production in each location l :

$$Y_l = A_l K_l^\alpha L_l^{1-\alpha}$$

- competitive input markets:

$$w_l = (1 - \alpha) A_l (K_l / L_l)^\alpha$$

$$r + \delta = \alpha A_l (K_l / L_l)^{\alpha-1}$$

- perfect capital mobility
 - r equalized across regions

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Equilibrium

1. HH optimization given prices
 - rural HHs solve V_R with $\gamma < 1$
 - urban HHs solve V_U with $\gamma = 1$
2. firm optimization
3. market clearing:
 - labor: $L_l = \int z_l d\mu$
 - capital: $K_R + K_U = \int a d\mu$
4. stationary distribution

ESTIMATION

ESTIMATION

ASSIGNED PARAMETERS

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

ESTIMATED PARAMETERS

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_{ϵ}	shock correlation
m_c	migration cost

$t = 6$ months

length of one model period

- maximizes available data moments for estimation

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$\beta = 0.98$
discount factor

$$V_{l \rightarrow l'} = \max_{a' \geq \underline{a}} u(c, R, x) \\ + \beta \mathbb{E}[V_{l'}(a', z'_R, z'_U, x', e', \gamma)]$$

- standard for 6-month period length (from 0.96 for annual)

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$$\sigma = 2$$

coefficient of risk aversion in CRRA utility

$$u(c, l, x) = \frac{\tilde{c}^{1-\sigma} - 1}{1 - \sigma}$$

- standard

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$$\alpha = 0.64$$

capital share in production

$$Y_l = A_l K_l^{\alpha} L_l^{1-\alpha}$$

- labor share: 0.36
- **data:** ILOSTAT

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$$\delta = 0.051$$

capital depreciation rate

$$r + \delta = \alpha A_l (K_l / L_l)^{\alpha-1}$$

- standard for 6-month period length (from 0.10 for annual)

ESTIMATION

ASSIGNED PARAMETERS

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

ESTIMATED PARAMETERS

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_{ϵ}	shock correlation
m_c	migration cost

$$A_R = 1$$

rural total factor productivity

$$Y_R = A_R K_R^{\alpha} L_R^{1-\alpha}$$

- normalized

ESTIMATION

ASSIGNED PARAMETERS

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

ESTIMATED PARAMETERS

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_{ϵ}	shock correlation
m_c	migration cost

$\lambda_s = 0.06$
urban job separation rate

$$\Pr(e' = 0 \mid e = 1, l = U) = \lambda_s, \quad \Pr(e' = 1 \mid e = 0, l = U) = \lambda_f$$

- **target moment:** 6-month unemployment hazard rate for urban migrants
- **data:** survey in *Barnett-Howell et al. (2025)*

ESTIMATION

ASSIGNED PARAMETERS

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

ESTIMATED PARAMETERS

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_{ϵ}	shock correlation
m_c	migration cost

$\lambda_f = 0.42$
urban job finding rate

$$\Pr(e' = 0 \mid e = 1, l = U) = \lambda_s, \quad \Pr(e' = 1 \mid e = 0, l = U) = \lambda_f$$

- **target moment:** 6-month employment hazard rate for urban migrants
- **data:** survey in *Barnett-Howell et al. (2025)*

ESTIMATION

ASSIGNED PARAMETERS

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_U	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

ESTIMATED PARAMETERS

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_{ϵ}	shock correlation
m_c	migration cost

$$m_U = 0.34$$

consumption-equivalent utility cost of migration

$$\tilde{c} = \frac{c}{1 + m_U \cdot \mathbf{1}(l = U, x = 0)}$$

- **target moment:** share of income inexperienced migrants are willing to give up to move back to rural
 - survey asked migrants what rural income would make them willing to move back
- **data:** survey in *Barnett-Howell et al. (2025)*

ESTIMATION

ASSIGNED PARAMETERS

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

ESTIMATED PARAMETERS

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_{ϵ}	shock correlation
m_c	migration cost

$$\pi_x = 0.45$$

urban experience gaining rate

$$\Pr(x' = 1 \mid l = U, x = 0) = \pi_x, \quad \Pr(x' = 0 \mid l = R, x = 1) = \pi_{-x}$$

- **target moment:** among rural households with rural preference, what share switches to urban preference within 6 months after migrating
 - location preference: require higher income to move to the other location
- **data:** survey in *Barnett-Howell et al. (2025)*

ESTIMATION

ASSIGNED PARAMETERS

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

ESTIMATED PARAMETERS

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_{ϵ}	shock correlation
m_c	migration cost

$$\pi_{-x} = 0.68$$

urban experience losing rate

$$\Pr(x' = 1 \mid l = U, x = 0) = \pi_x, \quad \Pr(x' = 0 \mid l = R, x = 1) = \pi_{-x}$$

- **target moment:** among urban households with urban preference, what share switches to rural preference within 6 months after migrating
 - location preference: require higher income to move to the other location
- **data:** survey in *Barnett-Howell et al. (2025)*

ESTIMATION

ASSIGNED PARAMETERS

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
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b	UI benefit
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ESTIMATED PARAMETERS

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_{ϵ}	shock correlation
m_c	migration cost

$$\gamma = 0.56$$

understatement of urban income by rural households

$$\tilde{w}_U = \gamma w_U$$

- **target moment:** avg expected Nairobi income by rurals as a share of actual Nairobi income, conditional on demographics
- **data:** *Barnett-Howell et al. (2025)*

ESTIMATION

ASSIGNED PARAMETERS

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
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ESTIMATED PARAMETERS

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_{ϵ}	shock correlation
m_c	migration cost

$$b = 0$$

urban unemployment benefit

$$y(e, U, z_R, z_U) = \begin{cases} w_U z_U & e = 1 \\ b & e = 0 \end{cases}$$

- no formal unemployment insurance in Kenya at baseline

ESTIMATION

ASSIGNED PARAMETERS

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_U	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

ESTIMATED PARAMETERS

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_{ϵ}	shock correlation
m_c	migration cost

$$\underline{a} = 0$$

borrowing limit

$$c + a' + \mathbf{1}(l = R, l' = U) \cdot m_c = (1 + r)a + y(e, l, z_R, z_U), \quad a' \geq \underline{a}$$

- no borrowing allowed at baseline

ESTIMATION

ASSIGNED PARAMETERS

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
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b	UI benefit
$\underline{a}$	borrowing limit

ESTIMATED PARAMETERS

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_{ϵ}	shock correlation
m_c	migration cost

$$A_U = 1.77$$

urban total factor productivity

$$Y_U = A_U K_U^{\alpha} L_U^{1-\alpha}$$

- **target moment:** urban–rural ratio of median incomes
- **data:** Kenya Integrated Household Budget Survey (KIHBS)

ESTIMATION

ASSIGNED PARAMETERS

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

ESTIMATED PARAMETERS

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$$\rho_R = 0.64$$

persistence of rural productivity

$$\log Z'_R = \rho_R \log Z_R + \epsilon'_R$$

$$\log Z'_U = \rho_U \log Z_U + \epsilon'_U$$

$$SD(\epsilon'_R) = \sigma_{\epsilon,R}, \quad SD(\epsilon'_U) = \sigma_{\epsilon,U}$$

$$\text{Corr}(\epsilon'_R, \epsilon'_U) = \rho_\epsilon$$

- **target moment:** standard deviation of log rural income
- **data:** Kenya Integrated Household Budget Survey (KIHBS)

ESTIMATION

ASSIGNED PARAMETERS

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

ESTIMATED PARAMETERS

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$$\rho_U = 0.78$$

persistence of urban productivity

$$\log Z'_R = \rho_R \log Z_R + \epsilon'_R$$

$$\log Z'_U = \rho_U \log Z_U + \epsilon'_U$$

$$SD(\epsilon'_R) = \sigma_{\epsilon,R}, \quad SD(\epsilon'_U) = \sigma_{\epsilon,U}$$

$$\text{Corr}(\epsilon'_R, \epsilon'_U) = \rho_\epsilon$$

- **target moment:** standard deviation of log urban income
- **data:** Kenya Integrated Household Budget Survey (KIHBS)

ESTIMATION

ASSIGNED PARAMETERS

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

ESTIMATED PARAMETERS

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$$\sigma_{\epsilon,R} = 1.16$$

volatility of rural productivity shocks

$$\log Z'_R = \rho_R \log Z_R + \epsilon'_R$$

$$\log Z'_U = \rho_U \log Z_U + \epsilon'_U$$

$$SD(\epsilon'_R) = \sigma_{\epsilon,R}, \quad SD(\epsilon'_U) = \sigma_{\epsilon,U}$$

$$\text{Corr}(\epsilon'_R, \epsilon'_U) = \rho_\epsilon$$

- **target moment:** standard deviation of residual rural log income after controlling for persistence
- **data:** survey in *Barnett-Howell et al. (2025)*

ESTIMATION

ASSIGNED PARAMETERS

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

ESTIMATED PARAMETERS

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$$\sigma_{\epsilon,U} = 1.07$$

volatility of urban productivity shocks

$$\log Z'_R = \rho_R \log Z_R + \epsilon'_R$$

$$\log Z'_U = \rho_U \log Z_U + \epsilon'_U$$

$$SD(\epsilon'_R) = \sigma_{\epsilon,R}, \quad SD(\epsilon'_U) = \sigma_{\epsilon,U}$$

$$\text{Corr}(\epsilon'_R, \epsilon'_U) = \rho_\epsilon$$

- **target moment:** standard deviation of residual urban log income after controlling for persistence
- **data:** survey in *Barnett-Howell et al. (2025)*

ESTIMATION

ASSIGNED PARAMETERS

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

ESTIMATED PARAMETERS

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$$\rho_\epsilon = 0.03$$

cross-location correlation of productivity shocks within a household.

$$\log Z'_R = \rho_R \log Z_R + \epsilon'_R$$

$$\log Z'_U = \rho_U \log Z_U + \epsilon'_U$$

$$SD(\epsilon'_R) = \sigma_{\epsilon,R}, \quad SD(\epsilon'_U) = \sigma_{\epsilon,U}$$

$$\text{Corr}(\epsilon'_R, \epsilon'_U) = \rho_\epsilon$$

- **target moment:** slope from regressing urban log income on rural log income for similar households (clustered by observables)
- **model moment:** slope from regressing potential urban log income on potential rural log income within household
- **data:** *Barnett-Howell et al. (2025)*

ESTIMATION

ASSIGNED PARAMETERS

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

ESTIMATED PARAMETERS

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_{ϵ}	shock correlation
m_c	migration cost

$$m_c = 1721$$

monetary migration cost

$$c + a' + 1(l = R, l' = U) m_c = (1 + r)a + y(e, l, z_R, z_U)$$

- **target moment:** share of Kenyan population living in cities
- **model moment:** share of households in urban region
- **data:** Kenya National Bureau of Statistics

ESTIMATED PARAMETERS AND TARGET MOMENTS

Parameter	Value	Target moment	Moment value	
			Data	Model
A_U	1.77	Urban–rural ratio of median incomes	5.15	4.95
ρ_R	0.64	Standard deviation of log rural income	1.49	1.55
ρ_U	0.78	Standard deviation of log urban income	1.57	1.60
$\sigma_{\epsilon,R}$	1.16	Standard deviation of residual rural log income	1.14	1.19
$\sigma_{\epsilon,U}$	1.07	Standard deviation of residual urban log income	1.05	1.10
ρ_{ϵ}	0.03	Slope of urban vs rural log incomes	0.065	0.068
m_C	1721	Urban population share	0.320	0.328

REPRODUCING EXPERIMENTS

CASH TRANSFER: EGGER ET AL. (2022)

SAMPLING

RCT

- 1 rural Kenyan county
- half of villages treated
- eligibility: thatched roof (1/3 of HHs)

MODEL

- eligibility: poorest 1/3 of rural HHs by assets
- randomly treat half

CASH TRANSFER: EGGER ET AL. (2022)

SAMPLING	TREATMENT
RCT • 1 rural Kenyan county • half of villages treated • eligibility: thatched roof (1/3 of HHs)	RCT • one-time unconditional transfer • 75% of mean annual expenditure
MODEL • eligibility: poorest 1/3 of rural HHs by assets • randomly treat half	MODEL • one-time asset transfer • 75% of mean rural annual consumption

CASH TRANSFER: EGGER ET AL. (2022)

SAMPLING	TREATMENT	MEASUREMENT
RCT • 1 rural Kenyan county • half of villages treated • eligibility: thatched roof (1/3 of HHs)	RCT • one-time unconditional transfer • 75% of mean annual expenditure	RCT • endline: 19 months after transfer • migration and household expenditure, treatment vs control
MODEL • eligibility: poorest 1/3 of rural HHs by assets • randomly treat half	MODEL • one-time asset transfer • 75% of mean rural annual consumption	MODEL • endline: 3 periods (18 months) • migration: urban residence at endline • consumption

CASH TRANSFER: EGGER ET AL. (2022) RESULTS

treatment arm	MIGRATION RATE CHANGE		CONSUMPTION CHANGE	
	RCT	MODEL	RCT	MODEL
cash transfer	0.0 pp	0.0 pp	11.6%	13.7%

- model matches negligible short-run migration response, magnitude of consumption response

UNEMPLOYMENT INSURANCE AND CASH TRANSFER: MINER (2025)

SAMPLING

RCT

- male rural workers aged 18–30 in 1 rural county
- most had prior migration experience
- sampled 11.7% of HHs per village

MODEL

- sample 11.7% of rural HHs
- prioritize experienced rural HHs

UNEMPLOYMENT INSURANCE AND CASH TRANSFER: MINER (2025)

SAMPLING	TREATMENT
RCT • male rural workers aged 18–30 in 1 rural county • most had prior migration experience • sampled 11.7% of HHs per village	RCT • UI: up to 15 days at average urban wage (60 USD) • cash: one-time 50 USD transfer
MODEL • sample 11.7% of rural HHs • prioritize experienced rural HHs	MODEL • UI: <i>b</i> matching 15 days of avg urban wage • cash: asset transfer, 50/60 of UI amount

UNEMPLOYMENT INSURANCE AND CASH TRANSFER: MINER (2025)

SAMPLING	TREATMENT	MEASUREMENT
RCT • male rural workers aged 18–30 in 1 rural county • most had prior migration experience • sampled 11.7% of HHs per village MODEL • sample 11.7% of rural HHs • prioritize experienced rural HHs	RCT • UI: up to 15 days at average urban wage (60 USD) • cash: one-time 50 USD transfer MODEL • UI: <i>b</i> matching 15 days of avg urban wage • cash: asset transfer, 50/60 of UI amount	RCT • endline: about 6 months after treatment • Nairobi residence and monthly income MODEL • endline: 1 period (6 months) • urban residence and income changes relative to rural baseline income

UNEMPLOYMENT INSURANCE AND CASH TRANSFER: MINER (2025) RESULTS

	MIGRATION RATE CHANGE		INCOME CHANGE	
	RCT	MODEL	RCT	MODEL
treatment arm				
unemployment insurance	10.9 pp	10.8 pp	0.0%	120.5%

- UI: model matches migration response, vastly overpredicts income response

UNEMPLOYMENT INSURANCE AND CASH TRANSFER: MINER (2025) RESULTS

treatment arm	MIGRATION RATE CHANGE		INCOME CHANGE	
	RCT	MODEL	RCT	MODEL
unemployment insurance	10.9 pp	10.8 pp	0.0%	120.5%
cash transfer	4.5 pp	3.0 pp	0.0%	9.1%

- UI: model matches migration response, vastly overpredicts income response
- cash: model matches migration response, overpredicts income response
 - migration response exceeds that in *Egger et al. (2022)* due to sample of experienced migrants

INFORMATION PROVISION: BARNETT-HOWELL ET AL. (2025)

SAMPLING

RCT

- 17,000 rural HHs in 5 counties
- 30% of HHs within sampled villages treated

MODEL

- randomly treat 30% of rural HHs

INFORMATION PROVISION: BARNETT-HOWELL ET AL. (2025)

SAMPLING	TREATMENT
RCT • 17,000 rural HHs in 5 counties • 30% of HHs within sampled villages treated	RCT • information: beliefs $\uparrow$ 4% • mentor: beliefs $\uparrow$ 10% • group: beliefs $\uparrow$ 12%
MODEL • randomly treat 30% of rural HHs	MODEL • information: $\gamma_{treated} = 1.04\gamma$ • mentor: $\gamma_{treated} = 1.10\gamma$ • group: experienced treated HHs set to $\gamma = 1$; inexperienced γ chosen to match average belief gain

INFORMATION PROVISION: BARNETT-HOWELL ET AL. (2025)

SAMPLING	TREATMENT	MEASUREMENT
RCT • 17,000 rural HHs in 5 counties • 30% of HHs within sampled villages treated MODEL • randomly treat 30% of rural HHs	RCT • information: beliefs $\uparrow$ 4% • mentor: beliefs $\uparrow$ 10% • group: beliefs $\uparrow$ 12% MODEL • information: $\gamma_{treated} = 1.04\gamma$ • mentor: $\gamma_{treated} = 1.10\gamma$ • group: experienced treated HHs set to $\gamma = 1$; inexperienced γ chosen to match average belief gain	RCT • endline: about 16 months after treatment • any Nairobi migrant and HH income MODEL • endline: 3 periods (18 months) • urban residence and income changes relative to rural baseline income

INFORMATION PROVISION: RESULTS

treatment arm	MIGRATION RATE CHANGE		INCOME CHANGE	
	RCT	MODEL	RCT	MODEL
information	1.8 pp	0.8 pp	6.7%	3.1%

- information: model underpredicts migration and income responses

INFORMATION PROVISION: RESULTS

treatment arm	MIGRATION RATE CHANGE		INCOME CHANGE	
	RCT	MODEL	RCT	MODEL
information	1.8 pp	0.8 pp	6.7%	3.1%
mentor	2.4 pp	2.4 pp	7.7%	7.1%

- information: model underpredicts migration and income responses
- mentor: model matches responses well
 - information arm has to be worse because of a smaller impact on beliefs

INFORMATION PROVISION: RESULTS

treatment arm	MIGRATION RATE CHANGE		INCOME CHANGE	
	RCT	MODEL	RCT	MODEL
information	1.8 pp	0.8 pp	6.7%	3.1%
mentor	2.4 pp	2.4 pp	7.7%	7.1%
group	2.2 pp	2.4 pp	0.0%	6.9%

- information: model underpredicts migration and income responses
- mentor: model matches responses well
 - information arm has to be worse because of a smaller impact on beliefs
- group: model matches migration response, overpredicts income response
 - income response smaller than mentor due to selection of experienced, low-return migrants
 - but difference between mentor and group is tiny in the model

QUANTITATIVE RESULTS

QUANTITATIVE EXERCISES

- track **long-run** treatment effects beyond each RCT endpoint

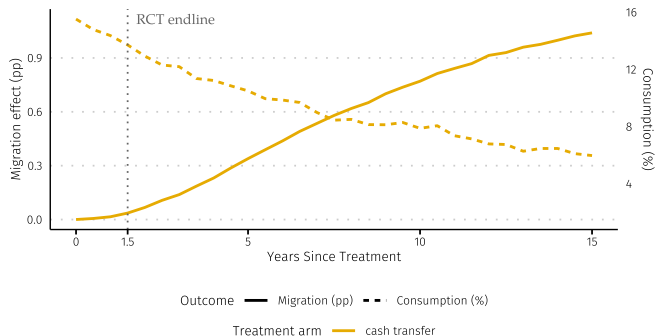
QUANTITATIVE EXERCISES

- track **long-run** treatment effects beyond each RCT endline
- roll out interventions **nationally** and re-solve general equilibrium

QUANTITATIVE EXERCISES

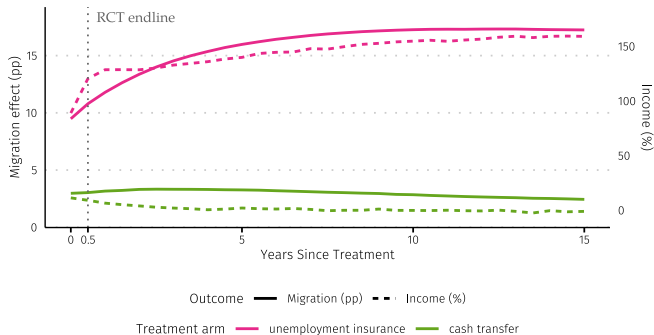
- track **long-run** treatment effects beyond each RCT endline
- roll out interventions **nationally** and re-solve general equilibrium
- shut down individual frictions to **decompose** the urban–rural income gap

LONG-RUN EFFECTS: CASH TRANSFER, EGGER ET AL. (2022)



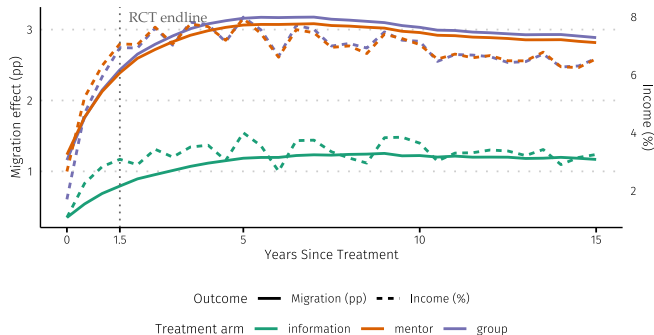
- consumption effects decay slowly as the transfer is spent
- migration is negligible at endline, but builds to 1 pp after 15 years
 - as more and more HHs experience shocks making migration attractive

LONG-RUN EFFECTS: UI AND CASH TRANSFER, MINER (2025)



- UI: 2/3 of effects realized by endline, continue to build
 - persistence partly reflects the model treatment: UI is available in every 6-month period
- cash: effects mostly realized by endline, then decay

LONG-RUN EFFECTS: INFORMATION PROVISION



- 3/4 of effects realized by endline, continue to build
- gains continue growing for a few years as favorable shocks make migration attractive
- treatment-control gaps later shrink: beliefs are identical when in the city

NATIONAL ROLLOUT

Scenario	Urban Share	URPG
Baseline	32.8%	4.945

NATIONAL ROLLOUT

Scenario	Urban Share	URPG
Baseline	32.8%	4.945
Barnett-Howell et al.: Individual	33.5%	4.940

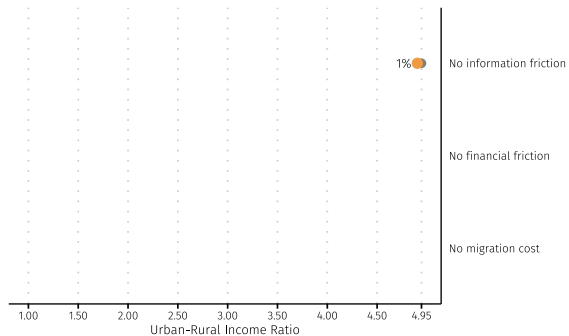
NATIONAL ROLLOUT

Scenario	Urban Share	URPG
Baseline	32.8%	4.945
Barnett-Howell et al.: Individual	33.5%	4.940
Barnett-Howell et al.: Mentor	35.6%	4.928

NATIONAL ROLLOUT

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Baseline	32.8%	4.945
Barnett-Howell et al.: Individual	33.5%	4.940
Barnett-Howell et al.: Mentor	35.6%	4.928
Miner: Unemployment insurance	58.1%	5.046

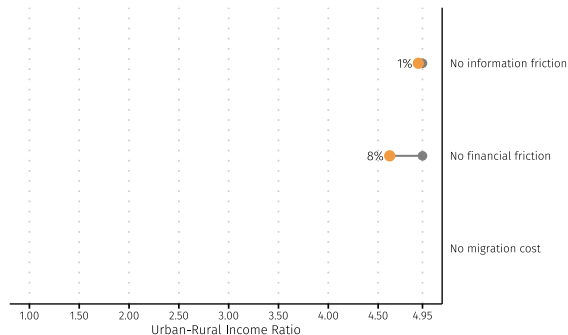
URBAN-RURAL INCOME GAP DECOMPOSITION



Scenario	Urban Share	URPG
Baseline	32.8%	4.945
No information friction	32.9%	4.904

- information explains little of URPG

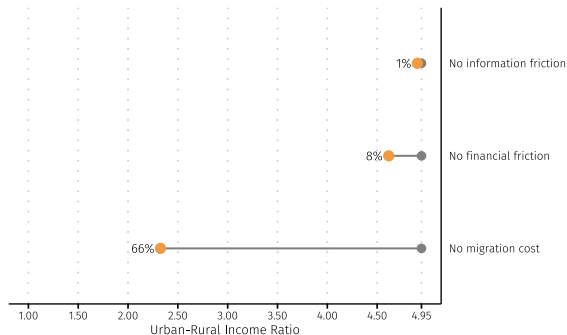
URBAN-RURAL INCOME GAP DECOMPOSITION



Scenario	Urban Share	URPG
Baseline	32.8%	4.945
No information friction	32.9%	4.904
No financial friction	1.2%	4.616

- information explains little of URPG
- removing borrowing constraint lowers the gap but collapses urban share
 - suggests that much of current migration is a form of self-insurance in the absence of formal credit markets

URBAN-RURAL INCOME GAP DECOMPOSITION



Scenario	Urban Share	URPG
Baseline	32.8%	4.945
No information friction	32.9%	4.904
No financial friction	1.2%	4.616
No migration cost	72.9%	2.325

- information explains little of URPG
- removing borrowing constraint lowers the gap but collapses urban share
 - suggests that much of current migration is a form of self-insurance in the absence of formal credit markets
- residual migration costs are the dominant barrier in the model

CONCLUSION

SUMMARY AND NEXT STEPS

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 - info frictions, borrowing constraints, urban risk, migration costs

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 - much of current migration may be overcoming fin. frictions, not spatial misallocation
 - most of the URP is unexplained by explicit frictions, picked up by residual costs
- **future steps:**
 - refine experiment mapping, estimation, simulation
 - alternative experiment mapping: calibrate treatment intensity to match treatment effects
 - non-linearities in firm production (slack) to reproduce local economic spillovers
 - cost-effectiveness comparison